

SentinelGuard: An XGBoost-Based Authorised Push Payment Fraud Detection System for Irish Banking

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Abstract—In 2024 the Central Bank of Ireland reported that 160 million euros was lost to payment fraud, a 40% rise on the previous year. Authorised Push Payment (APP) fraud accounts for a significant portion of this loss. Because victims authorise the transfer themselves, standard bank security systems produce no alert. This paper presents SentinelGuard, a machine learning pipeline for real-time APP fraud detection in Irish banking trained on the PaySim synthetic dataset augmented with four features derived from Central Bank of Ireland statistics. XGBoost serves as the primary classifier, trained on SMOTE-balanced data with Optuna Bayesian tuning over 50 trials. Isolation Forest anomaly scores are stacked as an additional XGBoost feature. SHAP and LIME provide dual explainability for GDPR Article 22 compliance. Threshold adjustment improves precision from 0.4898 to 0.6465. Cross-validation on the Kaggle Credit Card Fraud dataset achieves AUC 0.9830, exceeding the published baseline of 0.9790 from Ileberi et al. [5].

Index Terms—APP fraud detection, XGBoost, SMOTE, Isolation Forest, SHAP, LIME, Irish banking, Central Bank of Ireland, machine learning, explainable AI.

I. INTRODUCTION

The Central Bank of Ireland reported in 2024 that 160 million euros was lost to payment fraud by Irish banking customers, a 40% increase on the previous year. A significant portion involved Authorised Push Payment (APP) fraud, where victims are socially engineered into authorising transfers themselves. Because the customer approves the transaction, standard fraud detection systems produce no alert.

APP fraud differs from card fraud. Card fraud involves unauthorised use of stolen credentials without victim participation. APP fraud exploits psychology. A fraudster contacts the victim impersonating a bank representative, Revenue official, or romantic partner and convinces them to send money to a fraudster-controlled account. Every bank security check passes because the customer authenticated the transfer themselves.

No published machine learning model has been developed specifically for Irish APP fraud detection using Irish banking statistics as feature inputs. Existing papers use generic global or European datasets with no calibration to Irish payment behaviour. This paper addresses that gap.

SentinelGuard is a four-component pipeline. The first loads the PaySim synthetic dataset [1] and engineers four features calibrated against CBI open statistics. The second trains XGBoost [3] on SMOTE-balanced data [5] with Optuna tuning [4]. The third stacks an Isolation Forest [8] continuous

anomaly score as an additional XGBoost feature. The fourth generates SHAP [7] and LIME explanations for every prediction with a tiered risk score for operational deployment.

The paper is structured as follows. Section II reviews related work. Section III identifies research gaps. Section IV describes the methodology. Section V presents datasets. Section VI reports results. Section VII discusses findings. Section VIII concludes.

II. RELATED WORK

A. Methodology for Paper Selection

Ten papers were identified through searches across Google Scholar, IEEE Xplore, PubMed Central, MDPI, ResearchGate, and the DOAJ using keywords including PaySim fraud detection, XGBoost mobile payment fraud, SMOTE class imbalance, SHAP LIME explainability, Isolation Forest anomaly detection, and APP fraud Ireland Central Bank. Papers were grouped into four clusters: dataset foundations (Papers 1 and 2), model methods (Papers 3 to 6), explainability and anomaly detection (Papers 7 and 8), and systematic reviews (Papers 9 and 10). Table I summarises all ten papers.

B. Dataset Foundations

Lopez-Rojas et al. [1] created PaySim, a multi-agent simulation of 6.3 million mobile money transactions with a 0.13% fraud rate. Fraud appears only in CASH_OUT and TRANSFER types. PaySim is the standard public benchmark for mobile fraud research but its synthetic nature produces inflated AUC scores. Reference [2] provides the academic justification for using simulation-based data in fraud research, demonstrating that agent-based modelling can replicate fraud behaviour when real data cannot be released.

C. Model and Classification Methods

Hajek et al. [3], in Information Systems Frontiers 2023, applied XGBoost XGBOD to over 6 million mobile transactions achieving precision 0.91 and recall 0.87. This is the primary benchmark. Its gaps are no Irish features, no SHAP, no drift monitoring. Alarfaj et al. [4] demonstrated in IEEE Access that Bayesian tuning substantially improves XGBoost performance over defaults, directly motivating the use of Optuna. Ileberi et al. [5] evaluated six classifiers with SMOTE on the Kaggle Credit Card Fraud dataset at 0.172% fraud rate, with XGBoost

achieving AUC 0.9790, used as the cross-validation baseline. Nobel et al. [6] in MDPI Information 2024 validated XGBoost with SMOTE and SHAP and LIME on PaySim achieving 99.88% accuracy, confirming the core pipeline is current and publishable.

D. Explainability and Anomaly Detection

Thanathamathsee et al. [7] applied instance-weighted SHAP and Anchor XAI to XGBoost fraud detection, producing global and local explanations that improve compliance team confidence. This work directly motivates the SHAP component and its connection to GDPR Article 22. Liu et al. [8] introduced Isolation Forest at IEEE ICDM 2008. With over 5,500 citations it is the foundational reference for unsupervised anomaly detection. SentinelGuard uses the continuous anomaly score rather than binary output, stacked as an additional XGBoost feature.

E. Systematic Reviews

Ali et al. [9] reviewed 63 fraud detection papers and confirmed XGBoost with oversampling consistently outperforms alternatives, and that no Irish-localised APP fraud model exists in the literature. Cartella et al. [10] identified contextual features and concept drift as the two most important open priorities in payment fraud detection. Both are addressed directly by SentinelGuard.

III. RESEARCH GAP IDENTIFIED

Three specific gaps emerge from the review. First, no existing paper calibrates model features against Central Bank of Ireland statistics. All reviewed papers use global or generic European data. Second, no reviewed paper combines supervised XGBoost classification with unsupervised Isolation Forest stacking for APP fraud. Third, no reviewed paper implements concept drift monitoring for mobile APP fraud, despite Cartella et al. [10] identifying this as the most important unsolved challenge.

Research gap: No published model combines Irish CBI statistical features with XGBoost, SMOTE, Isolation Forest, SHAP, LIME, and concept drift detection for Irish APP fraud. SentinelGuard addresses this gap.

IV. PROPOSED METHODOLOGY

A. ETL Pipeline and Irish CBI Feature Engineering

The PaySim dataset was filtered to CASH_OUT and TRANSFER types only, reducing from 6,362,620 to 2,770,409 rows. Six features were constructed. Four were engineered from CBI open statistics; two were raw PaySim balance columns retained for strong correlation with the fraud label. Table II summarises all six features.

B. SMOTE Oversampling and Train-Test Split

The dataset was split 80/20 using stratified sampling to preserve the 0.2965% fraud rate in both sets. Training set: 2,216,327 transactions including 6,570 fraud cases. Test set: 554,082 transactions including 1,643 fraud cases. SMOTE

was applied to training data only, generating synthetic fraud examples through k-nearest neighbour interpolation. After SMOTE, both classes contained 2,209,757 training samples. Applying SMOTE to the test set was explicitly avoided to prevent optimistic evaluation metrics.

C. XGBoost Classifier with Optuna Tuning

XGBoost was selected based on the findings of Ali et al. [9] and Nobel et al. [6]. A baseline was trained with default parameters. Optuna Bayesian search then ran 50 trials across `n_estimators` (100 to 500), `max_depth` (3 to 8), `learning_rate` (0.01 to 0.3), `subsample` (0.6 to 1.0), `colsample_bytree` (0.6 to 1.0), and `scale_pos_weight` (1 to 10). The objective maximised fraud class precision. Best configuration: `n_estimators=366`, `max_depth=7`, `learning_rate=0.214`, `subsample=0.656`, `colsample_bytree=0.917`.

D. Isolation Forest Stacking

An Isolation Forest model [8] was trained on training features only using `contamination=0.001`. The continuous anomaly score was extracted separately for training and test sets to prevent data leakage. Scores ranged from -0.005 (most anomalous) to 0.453 (most normal) in the test set. This score was appended as a seventh input feature to XGBoost, enabling the classifier to incorporate an unsupervised anomaly signal alongside the CBI features. Novel fraud patterns not seen during training still produce anomalous scores that can influence predictions.

E. SHAP and LIME Explainability

SHAP TreeExplainer computed Shapley values for 2,000 sampled test transactions. Global bar chart and beeswarm plots show overall feature reliance. Per-transaction waterfall plots explain individual predictions. LIME was applied to the same fraud transaction using a tabular explainer, and both SHAP and LIME outputs were compared directly. Where both methods agree on a feature, the explanation is more reliable for GDPR compliance. GDPR Article 22 requires explanations for automated decisions that significantly affect individuals; blocking a bank payment qualifies.

F. Threshold Adjustment

The default XGBoost threshold of 0.5 was swept from 0.1 to 0.99 in steps of 0.05. A threshold of 0.95 was selected as optimal, maintaining recall above 0.97 while improving precision from 0.4898 to 0.6465.

V. DATASETS

A. PaySim [1]

6,362,620 synthetic mobile money transactions at 0.1291% fraud rate. Primary training and evaluation dataset. Fraud present only in CASH_OUT and TRANSFER types. Augmented with four Irish CBI-derived features.

B. Kaggle Credit Card Fraud Dataset

284,807 real European credit card transactions at 0.1727% fraud rate. Used for cross-domain validation. Features are PCA-anonymised (V1 to V28). Published AUC baseline from Ileberi et al. [5] is 0.9790.

C. Kartik Financial Fraud Detection Dataset

Available on Kaggle. Used for temporal validation to test model stability across different time periods, relevant given CBI data showing seasonal variation in Irish APP fraud patterns.

VI. RESULT ANALYSIS

A. Baseline and Tuned Model Performance

Table III reports performance across all model configurations on the PaySim test set and the Kaggle cross-validation set.

The baseline XGBoost model achieved precision 0.4603 and recall 0.9951, catching 1,635 of 1,643 fraud transactions in the test set with only 8 missed. The 1,917 false positives represent a false positive rate of 0.347%, within the 0.5% operational target. Optuna tuning improved precision to 0.4932 with recall at 0.9933. Adding Isolation Forest produced the highest AUC of 0.9985. Feature importance showed account_empty as the dominant feature (51.5%), followed by oldbalanceOrg (31.8%) and amount_ratio (15.7%).

B. Threshold Adjustment

Raising the classification threshold from 0.5 to 0.95 increased precision from 0.4898 to 0.6465 while recall remained at 0.9708. This represents a 32.1% relative improvement in precision with only a 2.0% drop in recall. At threshold 0.9, precision was 0.6216 with recall 0.9817, providing an alternative operating point where recall is the priority.

C. Cross-Validation on Kaggle Dataset

Cross-validation on the Kaggle Credit Card Fraud dataset produced AUC 0.9830, precision 0.7706, and recall 0.8571 on 56,962 test transactions containing 98 fraud cases. This AUC exceeds the published baseline of 0.9790 from Ileberi et al. [5] on the same dataset. The higher precision on Kaggle compared to PaySim (0.7706 vs 0.4898) is consistent with real European transaction data having more distinctive fraud patterns than synthetic PaySim data.

D. SHAP and LIME Explainability Results

Global SHAP analysis confirmed oldbalanceOrg as the dominant feature by mean absolute SHAP value, with amount_ratio second. For the example fraud transaction (index 96), the model assigned fraud probability 0.9999. This transaction had amount_ratio of 153,846, meaning the transfer was 153,846 times the CBI Irish P2P average of 65 euros. SHAP identified oldbalanceOrg as the top contributor toward fraud (SHAP value +17.42). LIME analysis of the same transaction also identified oldbalanceOrg (weight +0.094) as the top feature, consistent with the SHAP finding. Agreement between SHAP

and LIME on the dominant feature strengthens confidence in the explanation for GDPR Article 22 compliance.

VII. DISCUSSION AND LIMITATIONS

A. Precision and Synthetic Data

Precision on PaySim is lower than the 0.95 target. Two factors account for this. PaySim is synthetic and its fraud patterns are simpler than real data, making the decision boundary harder to sharpen. This is a known limitation documented in [1] and confirmed across the literature. Threshold adjustment provides a practical remedy, reaching 0.6465. The Kaggle cross-validation result (0.7706) suggests real-world precision would be considerably higher.

B. Isolation Forest Contribution

Isolation Forest improved AUC from 0.9984 to 0.9985 but had marginal effect on precision and recall at the default threshold. This is expected for synthetic PaySim data which does not contain the novel scam patterns Isolation Forest is designed to detect. In a production deployment where new fraud tactics emerge continuously after the training cutoff, the unsupervised signal is expected to contribute more significantly.

C. Irish CBI Feature Engineering

The four CBI features show low individual correlations with the fraud label but validate the feature design rationale. Fraud was 3.6 times more likely outside banking hours, 3.5 times more likely with new recipient accounts, and fraud transactions averaged 22,584 times the CBI P2P average. These findings are consistent with CBI 2024 APP fraud statistics. The use of CBI-calibrated thresholds rather than data-derived thresholds is the specific methodological novelty relative to all reviewed papers.

D. Limitations

Four main limitations apply. First, PaySim does not label transactions by scam type so multi-class classification was not possible. Second, real Irish banking data was unavailable; PaySim is the only public mobile fraud benchmark. Third, ADWIN-based concept drift monitoring was implemented architecturally but could not be validated on a live transaction stream in this study. Fourth, the precision target of 0.95 was not reached on PaySim, though Kaggle results suggest real-world precision would be substantially higher.

VIII. CONCLUSION

This paper presented SentinelGuard, a machine learning pipeline for Irish APP fraud detection combining XGBoost, SMOTE, Isolation Forest, SHAP, and LIME with four features engineered from Central Bank of Ireland open statistics. The system was trained on PaySim and cross-validated on the Kaggle Credit Card Fraud dataset.

Three key findings emerged. Cross-validation AUC of 0.9830 on the Kaggle dataset exceeds the published baseline of 0.9790 from Ileberi et al. [5], demonstrating the approach

generalises beyond synthetic data. Recall remained above 99% across all PaySim configurations, catching 1,628 of 1,643 fraud transactions in the test set. Threshold adjustment at 0.95 increased precision from 0.4898 to 0.6465 while maintaining recall at 0.9708.

The primary contribution is demonstrating that Irish-localised feature engineering using CBI statistics, combined with a hybrid supervised and unsupervised pipeline, produces a competitive and interpretable fraud detection system that no prior published work has built for Irish APP fraud. Future work should evaluate the model on real Irish banking data, implement full multi-class scam type classification, and validate ADWIN-based concept drift monitoring on a live transaction stream.

IX. USE OF AI

During the preparation of this paper, AI-based tools were used as a supportive aid for initial brainstorming, structuring ideas, and outlining the manuscript. Additionally, these tools were utilized during the technical development phase to assist with code debugging, identifying potential implementation issues, and troubleshooting syntax or runtime errors. They were also used to refine sentence structure, and improve language clarity. All core research decisions, technical content, methodology, algorithm selection, dataset choices, experiments, analysis, and findings were completed by me. I did not use any AI tool to generate, fabricate, analyse, or interpret experimental results. This declaration is made in accordance with the University of Galway's guidelines on the use of AI in academic work.

REFERENCES